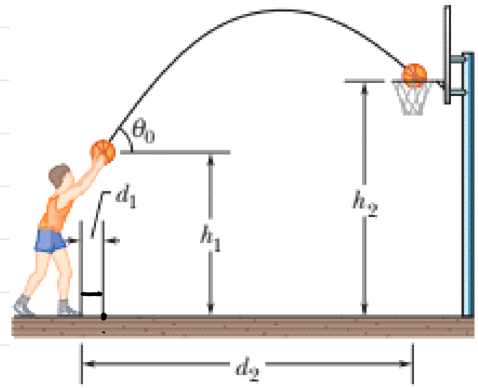
Principles of Physics-I, PHY111

H.W

Prob 5: (a) At what initial speed must the basketball player in Figure throw the ball, at angle 55° above the horizontal, to make the foul shot? The horizontal distances are d_1= 1.0 ft and d_2 = 14 ft, and the heights are h_1 = 7.0 ft and h_2 =10 ft.
(b) Calculate the flight of time of the ball.
(c) Calculate the velocity of the ball just before entering the net.



Soln: (a) $x = vt$

$\Rightarrow d_2 - d_1 = v_0 \cos\theta \, t$

$\Rightarrow 14 - 1 = v_0 \cos 55° \, t$

$\Rightarrow \boxed{t = \frac{13}{v_0 \cos 55°}}$ — (1)

Now, $y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2$

$\Rightarrow 10 = 7 + v_0 \sin 55° \times \frac{13}{v_0 \cos 55°} - 16 \times \left(\frac{13}{v_0 \cos 55°}\right)^2$ (Since 32 ft/s²

$\Rightarrow 3 = 18.56 - \frac{8243.90}{v_0^2}$

$\Rightarrow \boxed{v_0 = 23.01 \text{ ft/sec}}$ Ans

(b) Flight of time, $t = \frac{13}{v_0 \cos 55°} = \frac{13}{23.01 \times \cos 55°} = 0.985$ sec

$\boxed{t = 0.985 \text{ sec}}$ Ans

(c) $v_x = v_0 \cos\theta$

$\boxed{v_x = 13.19 \text{ ft/sec}}$ Ans

$v_y = v_{0y} - gt = 23.01 \times \sin 55 - 32 \times 0.985$

$= \boxed{v_y = -12.67 \text{ ft/s}}$

So, $\boxed{\vec{v} = 13.19\hat{i} - 12.67\hat{j}}$

Ans